

Day 04: Measures of Dispersion

Understanding Data Spread

100 Days of Data Science

Why Dispersion is Important

Measures of central tendency tell us the center of data, but they do not tell us how spread out the data values are.

Two datasets can have the same mean but very different variability.

Dispersion helps us understand:

- Consistency of data
- Risk and uncertainty
- How much values differ from the average

Main Measures of Dispersion

The common measures are:

- Range
- Variance
- Standard Deviation

Range

Range is the difference between the maximum and minimum values.

Formula

$$\text{Range} = \text{Max} - \text{Min}$$

Example

Data: 10, 20, 30, 40

$$\text{Range} = 40 - 10 = 30$$

Limitations

- Depends only on extreme values
- Highly affected by outliers

Variance

Variance measures the average squared deviation from the mean.

Formula (Sample Variance)

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

Intuition

Variance shows how far data points are spread from the average.

Why Squared?

- Prevents positive and negative values from canceling
- Gives more weight to larger deviations

Standard Deviation

Standard deviation is the square root of variance.

Formula

$$s = \sqrt{s^2}$$

Why it is Useful

- Same unit as original data
- Easy to interpret
- Most widely used in statistics and ML

Simple Example

Data: 2, 4, 6

Mean = 4

Variance = 4

Standard Deviation = 2

Interpretation: Values deviate about 2 units from the mean.

Role in Data Science

- Detecting outliers
- Understanding data consistency
- Feature scaling in ML
- Risk analysis

Common Mistakes

- Ignoring spread and looking only at mean
- Misinterpreting variance units

Final Takeaway

Central tendency shows the center. Dispersion shows the spread.

Good data analysis always looks at both.